MySQL Assignment 2 – Querying Data

DML comments

1. Distinct Values:

A query to retrieve distinct salaries from the Employees table.

Here this is my inserted values

```
133 •  select * from employees;
134
135
136
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
1	Ajay	M	23	2018-01-15	Developer	1	101	550000.00
2	sanjay	M	28	2016-03-09	HR Executive	2	102	68000.00
3	Viji	F	25	2020-12-25	Financial Analyst	3	103	53000.00
4	Sharma	F	27	2017-10-14	Marketing manager	4	104	69000.00
5	joseph	M	32	2015-05-25	Sales officer	5	105	550000.00
6	Priya	F	29	2021-02-18	Admin	6	106	46000.00
7	Riyan	M	26	2025-08-29	Production Lead	7	107	35000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```
40 •  select distinct salary from employees;
```

esult Grid | Filter Rows: | Export: | Wrap C

salary
550000.00
68000.00
53000.00
69000.00
46000.00
35000.00

2. Alias (AS):

Provide aliases for the "age" and "salary" columns as "Employee_Age" and

"Employee_Salary", respectively.

```
42 •  select age as Employee_Age from employees;
43 •  select salary as Employee_salary from employees;
44
45 •  select  employee_name,salary from employees where salary > 50000 a
46
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Age
23
28
25
27
32
29
26

```
43 •  select salary as Employee_salary from employees;
44
45
46
47
48 •  select  employee_name,salary from employees where salary >
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_salary
550000.00
68000.00
53000.00
69000.00
550000.00
46000.00
35000.00

Where Clause & Operators:

Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

Find the employee whose designation is missing and fill it with "Data Scientist".

```
47
48 •  select  employee_name,salary from employees where salary > 50000 and hire_date < '2016-01-01';
49
50
```

Filter Rows: | Export: | Wrap Cell Content:

employee_name	salary
joseph	550000.00

```
55
56 •  select * from employees where designation is null ;
57
58 •  update employees set designation ='data scientist' where designation = ' ';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

ORDER BY:

Find employees sorted by department ID in ascending order and salary in descending order.

```
51 •  select * from employees order by department_id asc ,salary desc;
52
```

esult Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
1	Ajay	M	23	2018-01-15	Developer	1	101	550000.00
2	sanjay	M	28	2016-03-09	HR Executive	2	102	68000.00
3	Viji	F	25	2020-12-25	Financial Analyst	3	103	53000.00
4	Sharma	F	27	2017-10-14	Marketing manager	4	104	69000.00
5	joseph	M	32	2015-05-25	Sales officer	5	105	550000.00
6	Priya	F	29	2021-02-18	Admin	6	106	46000.00
7	Riyan	M	26	2025-08-29	Production Lead	7	107	35000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

LIMIT:

Display the first 5 employees hired in the year 2018.

```
58 •  select * from employees where hire_date >= '2018-01-01' and hire_date < '2019-01-01' order by hire_date limit 5;
59
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
1	Ajay	M	23	2018-01-15	Developer	1	101	550000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Aggregate Functions:

Calculate the sum of all salaries in the Finance department.

```
54
55 •    select sum(salary) as  total_Finance from employees;
56
57
58
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_Finance
1371000.00

Find the minimum age among all employees.

```
60
61
62 •  select min(age) as minimum_Age from employees;
63
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

minimum_Age
23

GROUP BY: List the maximum salary for each location.

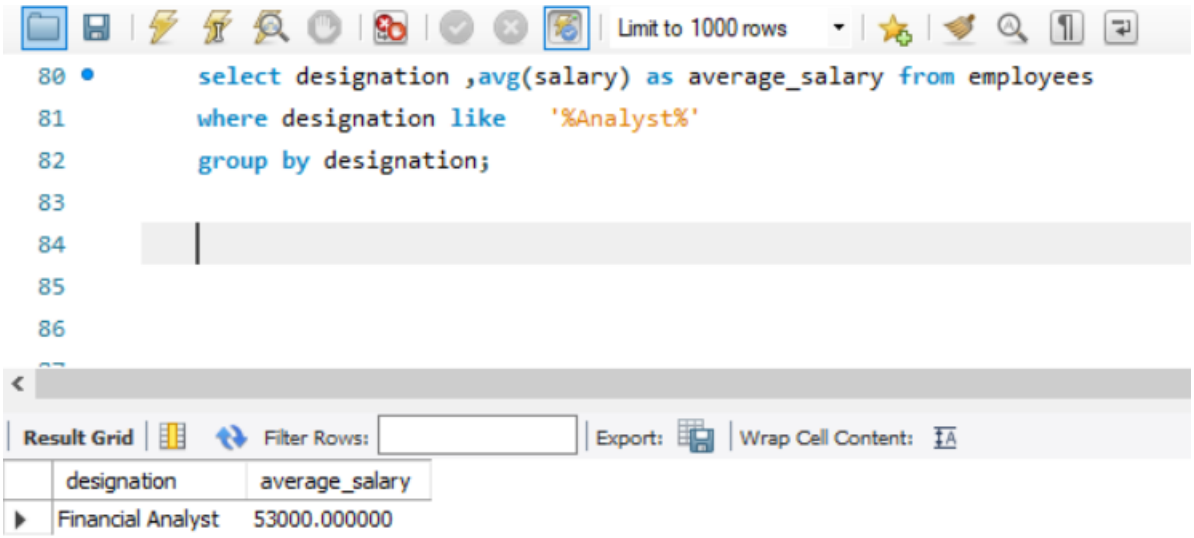
```
73 •  select
74    l.location_name, max(e.salary) as maximum_salary
75    from location1 l,
76    employees e
77     where  l.location_id=e.location_id
78    group by l.location_name;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

location_name	maximum_salary
chennai	550000.00
Banglore	68000.00
Salem	53000.00
Coimbatore	69000.00
Madurai	550000.00
Trichy	46000.00
Mumbai	35000.00

Calculate the average salary for each designation containing the word

'Analyst'.



HAVING:

Find departments with less than 3 employees.

```
84      ## having
85      select department_id,count(*) AS total_employees from employees
86      group by department_id
87      having count(*) < 3;
88
89
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_id	total_employees
1	1
2	1
3	1
4	1
5	1
6	1
7	1

Inner Join:

List employee names, their designations, and department names where

employees are assigned to a department.

```
90      select e.employee_name,e.designation,d.department_name
91      from employees e
92      inner join departments as d
93      on e.department_id=d.department_id;
94
95
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_name	designation	department_name
Ajay	Developer	IT
sanjay	HR Executive	HR
Viji	Financial Analyst	Finance
Sharma	Marketing manager	Marketing
joseph	Sales officer	Sales
Priya	Admin	Administration
Riyan	Production Lead	Production

Left Join: List all departments along with the total number of employees in each department, including departments with no employees.

```
100 •       select d.department_name,
101         count(e.employee_id)as
102         total_employees
103         from departments d
104         left join employees e on d.department_id= e.department_id
105         group by d.department_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_name	total_employees
IT	1
HR	1
Finance	1
Marketing	1
Sales	1
Administration	1
Production	1

Right Join:

Display all locations along with the names of employees assigned to each

location. If no employees are assigned to a location, display NULL for

employee name.

```
use Employees1_DB;



select l.location,e.employee_name
from employees e
right join location l
on e.location_id=l.location_id;
```